



Faculty of Engineering
Electrical Engineering Department

COMMUNICATIONS NETWORKS

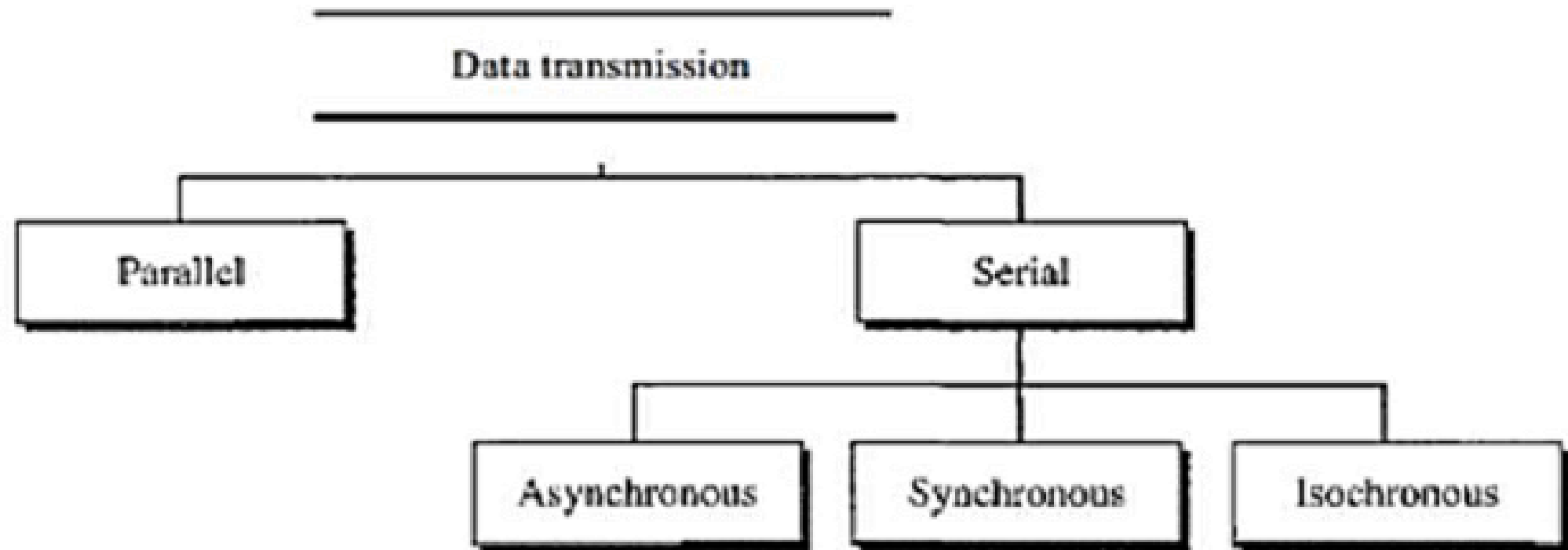
ELC-411

Lecture 3

Dr esraa ahmed el-refaay

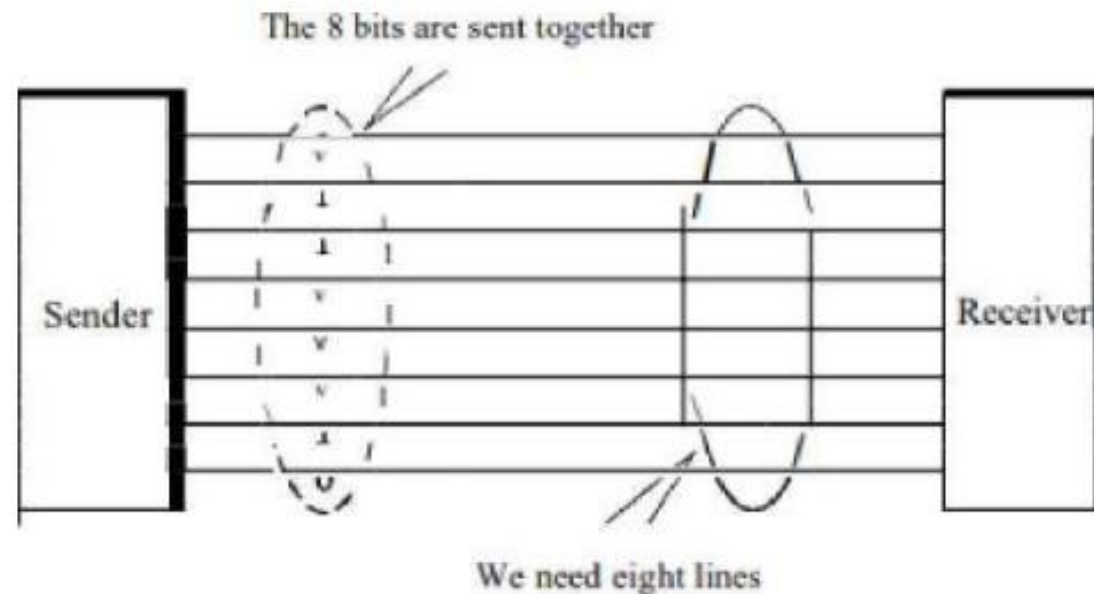
❑ Transmission modes

Data transmission and modes



❑ Transmission modes

Parallel Transmission: Use n wires to send n bits at one time. That way each bit has its own wire, and all n bits of one group can be transmitted with each clock tick from one device to another.

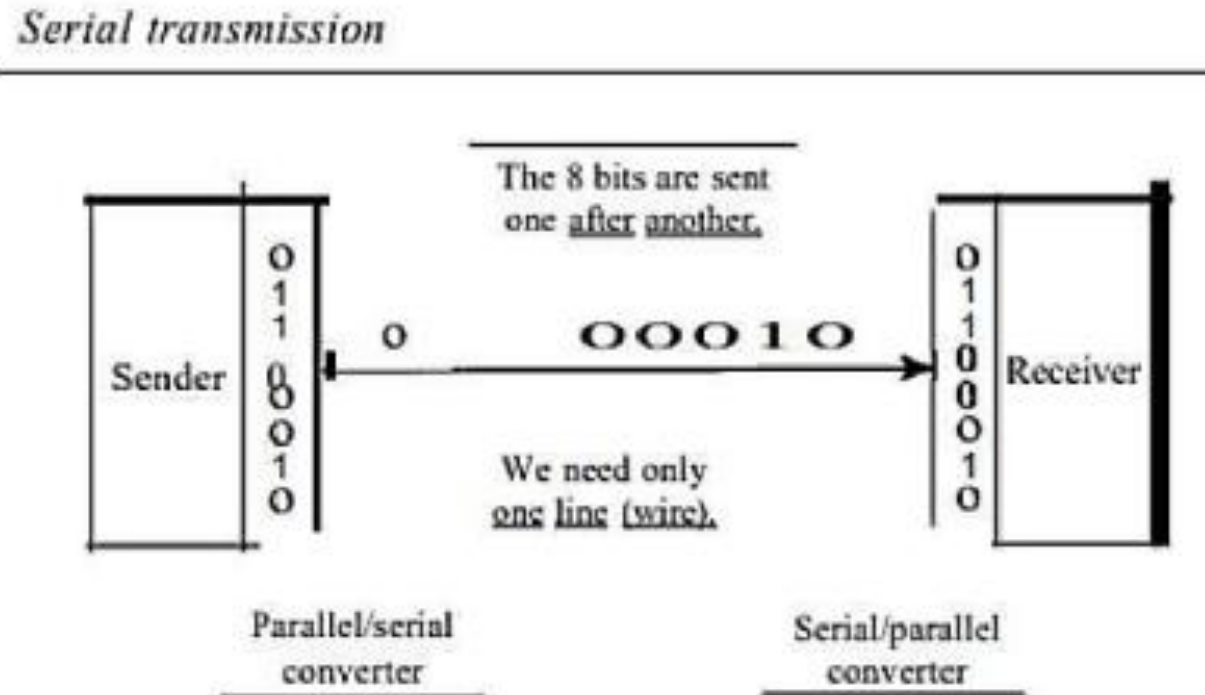


The advantage of parallel transmission **is speed**. All else being equal, parallel transmission can increase the transfer speed by a factor of n over serial transmission.

❑ Transmission modes

Serial Transmission

In serial transmission one bit follows another, so we need only one communication channel rather than n to transmit data between two communicating devices

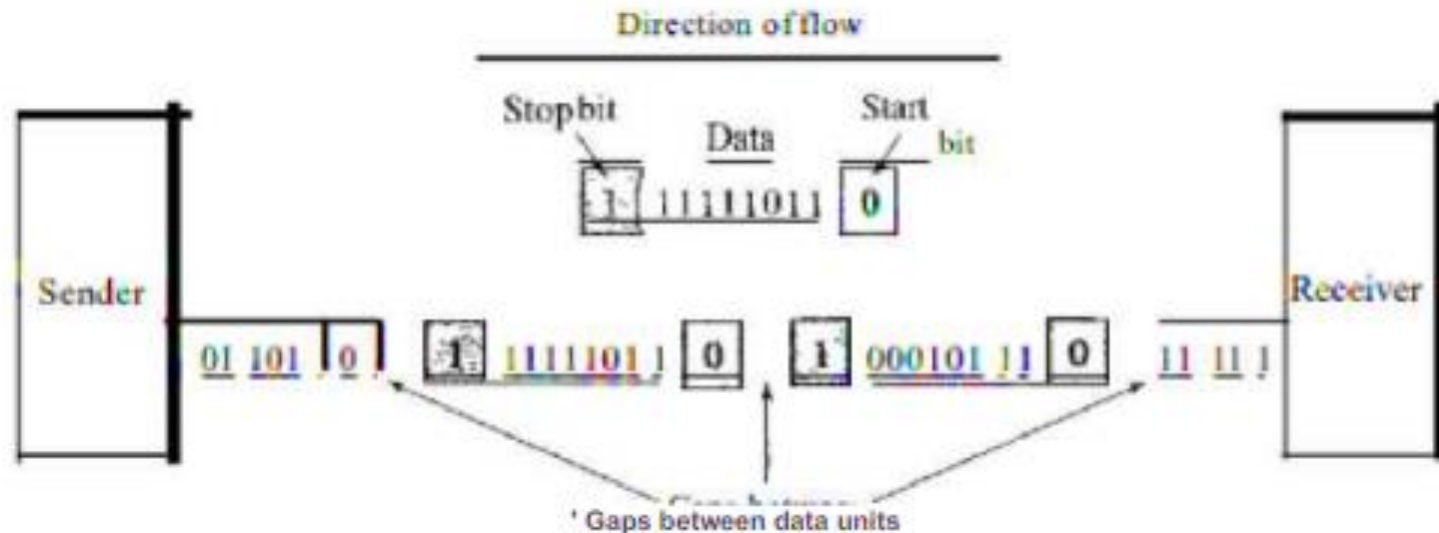


UART (Universal Asynchronous Receiver/Transmitter)

❑ Transmission modes

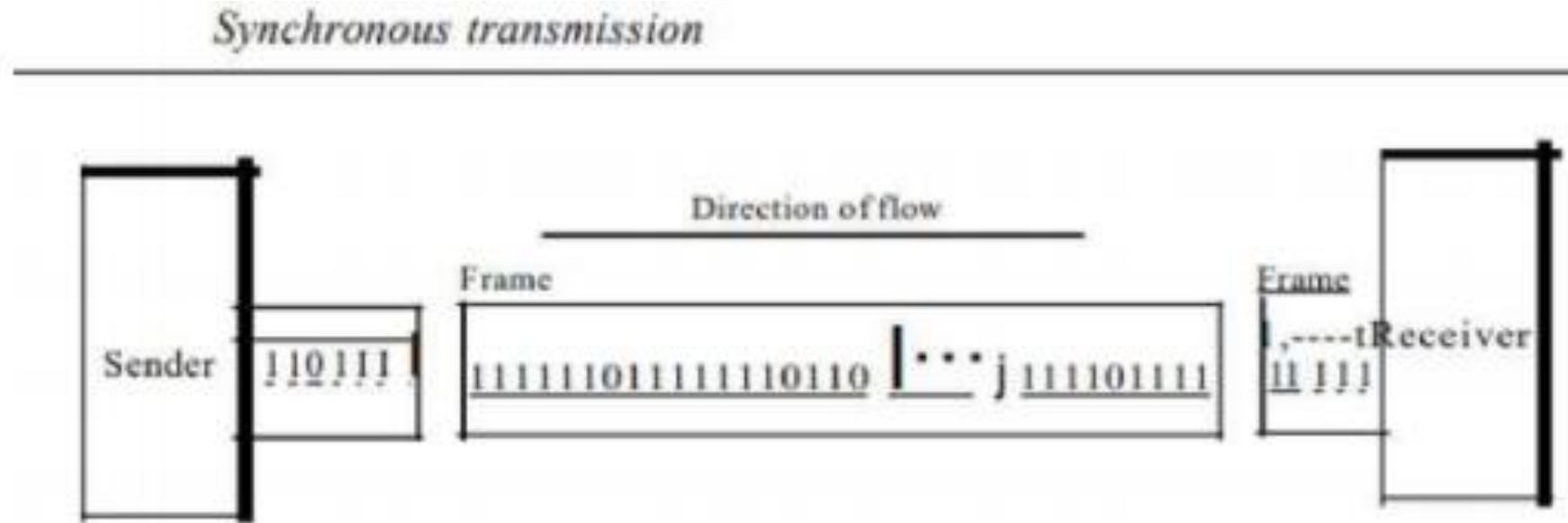
Serial transmission occurs in one of three ways: **asynchronous**, **synchronous**, and **isochronous**

Asynchronous Transmission: is so named because the timing of a signal is unimportant. Instead, information is received and translated by agreed upon patterns. As long as those patterns are followed, the receiving device can retrieve the information without regard in which it is sent. Patterns are based on grouping the bit stream into bytes. Each group, usually 8 bits, is sent along the link as a unit. The sending system handles each group independently, relaying it to the link whenever ready, without regard to a timer.



❑ Transmission modes

Synchronous Transmission : In synchronous transmission, the bit stream is combined into longer "frames," which may contain multiple bytes. Each byte, however, is introduced onto the transmission link without a gap between it and the next one. It is left to the receiver to separate the bit stream into bytes for decoding purposes.



- HDLC (High-Level Data Link Control)
- PPP (Point-to-Point Protocol)
- Ethernet
- USB

❑ Transmission modes

- Streaming Media .
- VoIP (Voice over IP).
- Video conference.

Isochronous transmission

In **real-time audio and video**, in which uneven delays between frames are *not acceptable*, synchronous transmission fails.

For example, TV images are broadcast at the rate of 30 images per second; they must be viewed at the same rate.

If each image is sent by using one or more frames, there should be no delays between frames. For this type of application, synchronization between characters is not enough; the entire stream of bits must be synchronized. The isochronous transmission guarantees that **the data arrive at a fixed rate.**

❑ Digital to digital encoding (Line Coding)

Data and information can be stored in 2 ways **Analog & Digital**.

To transmit data digitally, it need to be first converted to digital form.

Digital to Digital Encoding & Conversion means

converting digital data to digital signals.

The conversion can be done in 2 ways

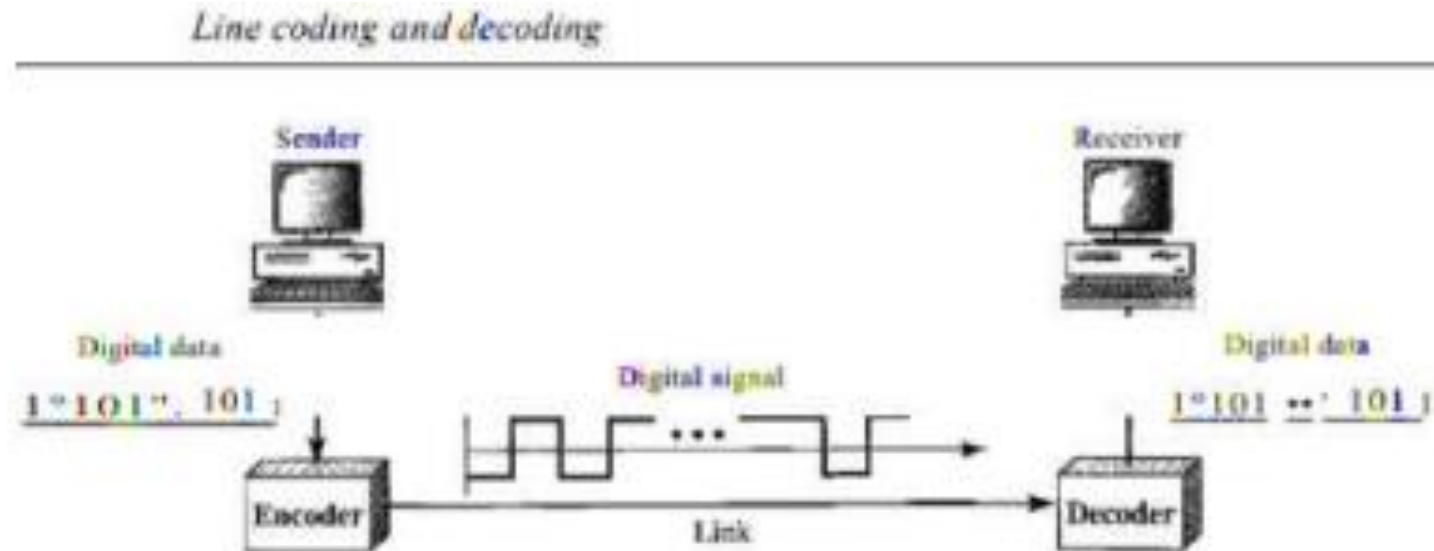
- **Line coding**
- **Block coding**

❑ Digital to digital encoding (Line Coding)

Line coding is the process of converting digital data to digital signals.

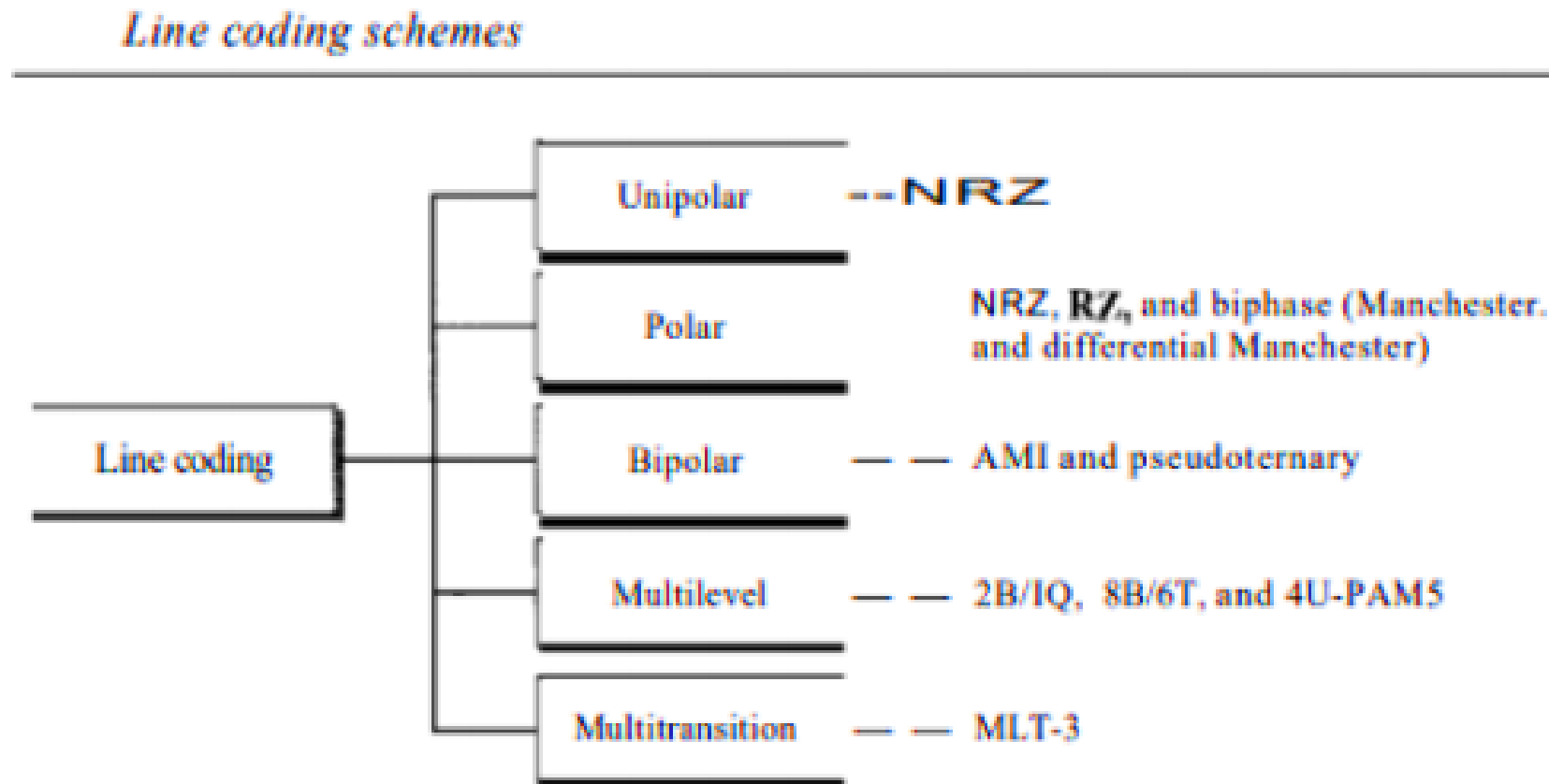
- Digital data- found in binary format 101001.
- Digital signals- is denoted by discrete signals.

We assume that data, in the form of text, numbers, graphical images, audio, or video, are stored in computer memory as sequences of bits. Line coding converts a sequence of bits to a digital signal. At the sender, digital data are encoded into a digital signal; at the receiver, the digital data are recreated by decoding the digital signal. Figure shows the process.



□ Line Coding Schemes

We can roughly divide line coding schemes into five broad categories

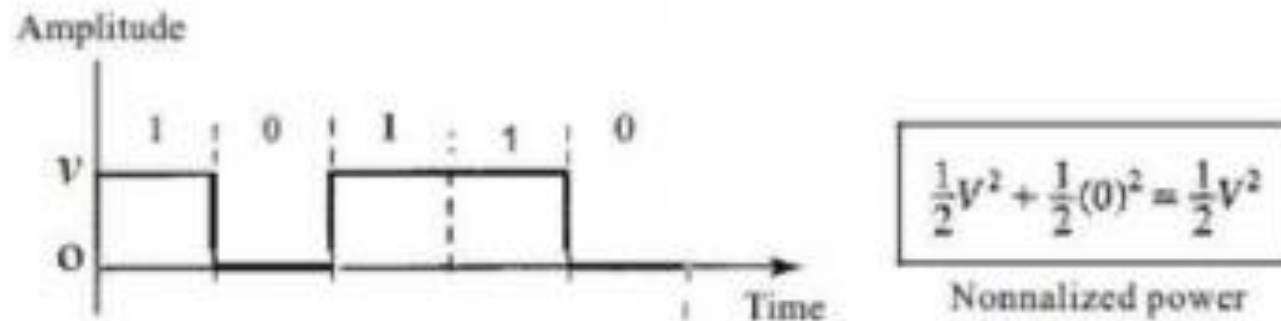


❑ Line Coding Schemes

Unipolar Scheme: all the signal levels are on **one side** of the time axis, either above or below.

NRZ (Non-Return-to-Zero): Traditionally, a unipolar scheme was designed as a non-return-to-zero (NRZ) scheme in which the **positive voltage defines bit 1** and the **zero voltage defines bit 0**. It is called NRZ because the signal does not return to zero at the middle of the bit.

Unipolar NRZ scheme



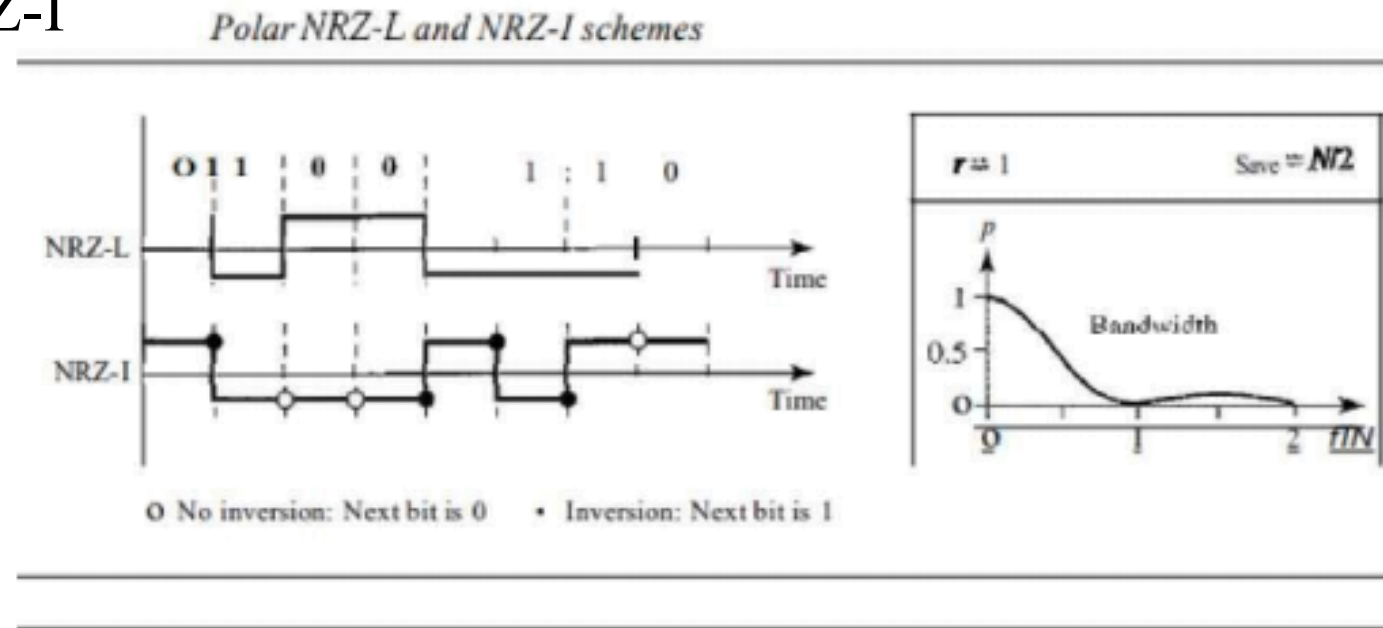
❑ Line Coding Schemes

Polar Schemes: uses multiple voltage levels to represent binary values.

the voltages are on the both sides of the time axis. For example, the voltage level for 0 can be positive and the voltage level for 1 can be negative.

✓ Non-Return-to-Zero (NRZ)

In polar NRZ encoding, we use two levels of voltage amplitude. We can have two versions of polar NRZ: NRZ-L and NRZ-I

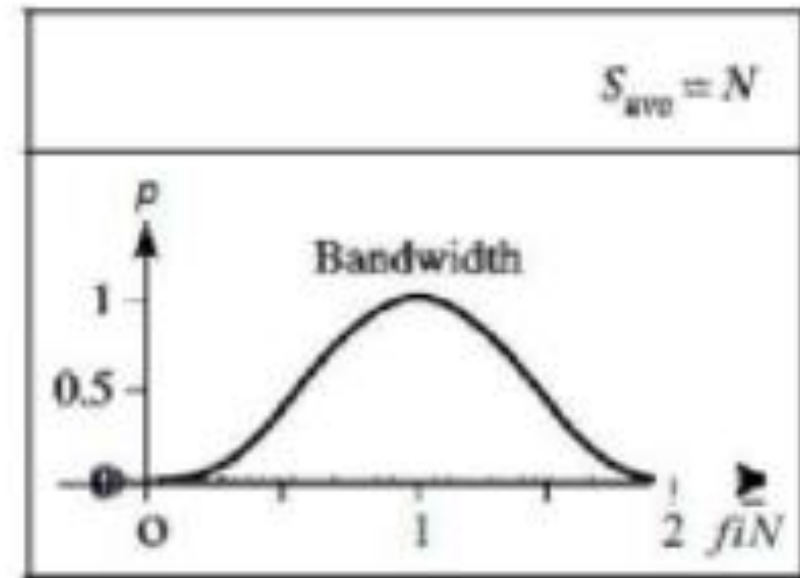
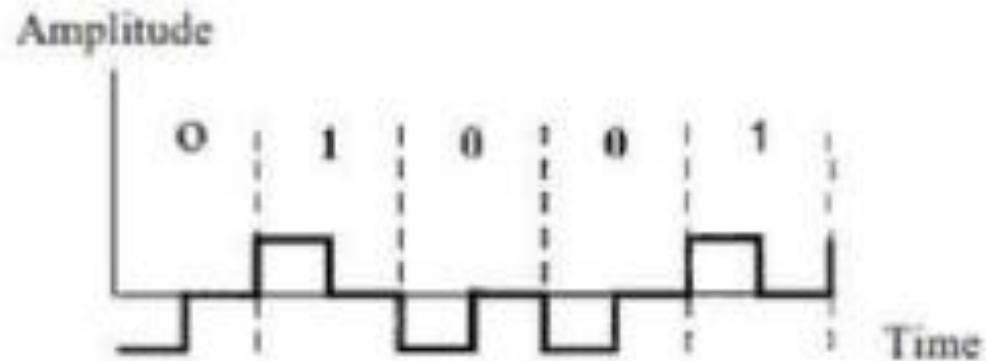


In NRZ-L the level of the voltage determines the value of the bit. In NRZ-I the inversion or the lack of inversion determines the value of the bit.

❑ Line Coding Schemes

✓ Return-to-Zero (RZ)

The main problem with NRZ encoding occurs when **the sender and receiver clocks are not synchronized**. The receiver does not know when one bit has ended and the next bit is starting. One solution is the return-to-zero (RZ) scheme, which uses three values: positive, negative, and zero. In RZ, the signal changes not between bits but during the bit.



□ Line Coding Schemes

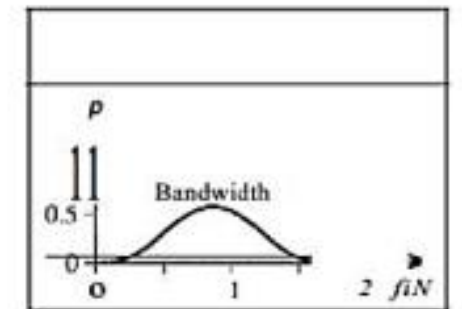
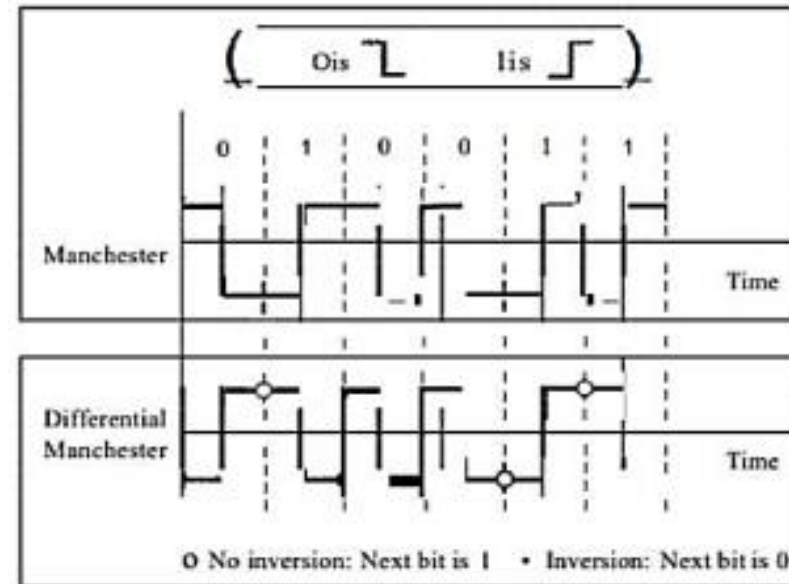
Bi phase: Manchester and Differential Manchester

The idea of RZ (transition at the middle of the bit) and the idea of NRZ-L are combined into the Manchester scheme.

In Manchester encoding, the duration of the bit is divided into two halves. The voltage remains at one level during the first half and moves to the other level in the second half. The transition at the middle of the bit provides synchronization.

Differential Manchester, on the other hand, combines the ideas of RZ and NRZI. There is always a transition at the middle of the bit, but the bit values are determined at the beginning of the bit.

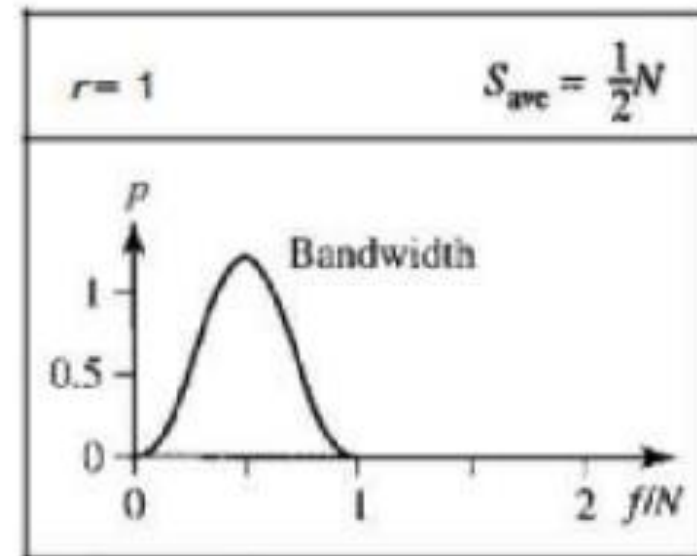
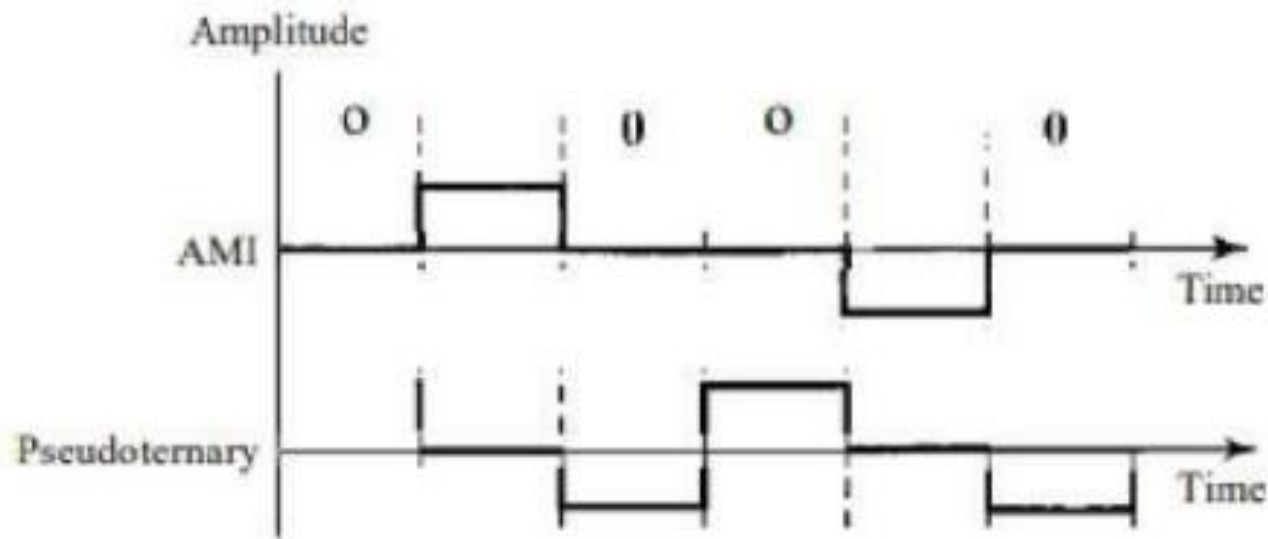
If the next bit is 0, **there is a transition**;
if the next bit is 1, **there is none**.



□ Line Coding Schemes

Bi polar: In bipolar encoding (sometimes called multilevel binary), there are three voltage levels: **positive, negative, and zero**. The voltage level for one data element is at zero, while the voltage level for the other element alternates between positive and Negative

Alternate Mark Inversion (AMI) and Pseudoternary Figure below shows two variations of bipolar encoding



❑ Line Coding Schemes

Multilevel Line Coding

Instead of using only **two voltage levels** (like $+V / -V$ or $0 / 1$), multilevel line coding uses **more than two levels** to represent data.

The main goal: **increase bandwidth efficiency**.

Characteristics:

1. More than two levels:

1. Each symbol can represent **more than 1 bit**.
2. Example:
 1. With 4 levels $\rightarrow$ each symbol = 2 bits.
 2. With 8 levels $\rightarrow$ each symbol = 3 bits.

2. Higher efficiency:

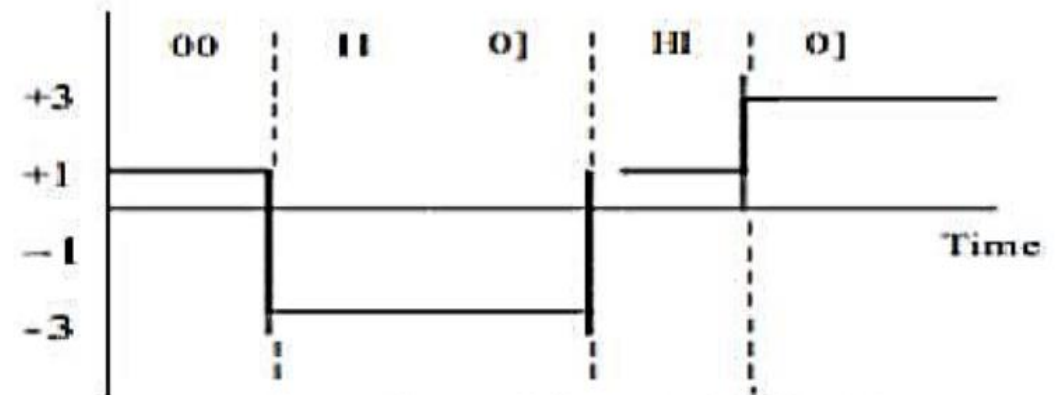
1. More bits per symbol $\rightarrow$ lower **baud rate** for the same data rate.
2. Saves bandwidth compared to binary schemes.

3. More complex:

1. Needs more complex transmitters and receivers.
2. More sensitive to **noise**, since levels are closer together.

Next bits	Previous level: positive	Previous level: negative
	Next level	Next level
00	+1	-1
01	+3	-3
10	-1	+1
11	-3	+3

Transition table



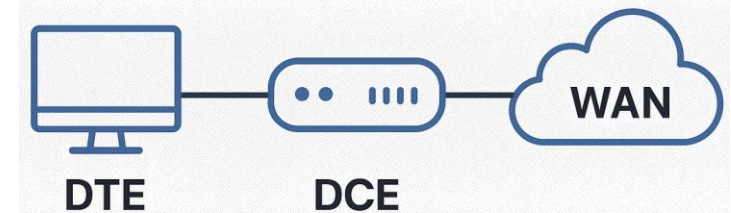
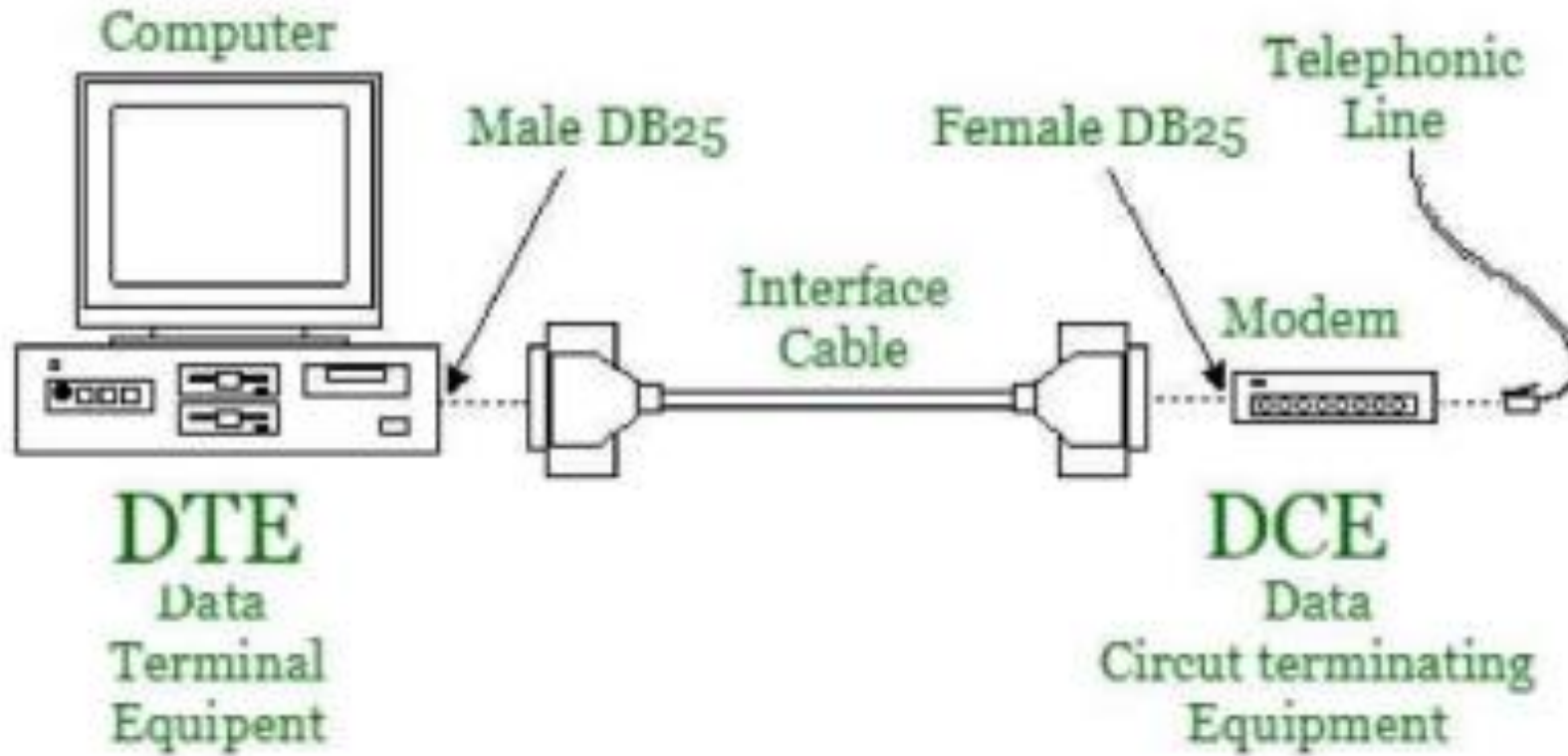
Assuming positive original level

2BIQ

□ Line Coding Schemes

Scheme	Example	Advantages	Disadvantages
Unipolar	NRZ	Simple implementation	Large DC component, poor sync
Polar NRZ-L/I	NRZ	Less DC than unipolar	Synchronization problem with long runs of 0s/1s
Polar RZ	RZ	Better synchronization	Requires double bandwidth
Manchester	Ethernet (10 Mbps)	Excellent self-synchronization	Consumes double bandwidth
Diff Manchester	Token Ring	Error detection, good sync	Higher bandwidth usage
Bipolar (AMI)	T1 lines	No DC component, good for voice	Requires more complex receivers
Multilevel	MLT-3, 2B1Q	Bandwidth efficient	More complex encoding/decoding

❑ DTE-DCE Interface



❑ Data Terminal Equipment (DTE)

At the physical layer, the function of DTE is to look for clocking from the device on the other end of the cable on a link and look for a frame-switching service.

The device receives clocking. The customer devices or the DTE pass data from a host computer on a customer network for transmission over the Wide Area Network (WAN).

Also, its function is to convert user information into signals or reconvert received signals. DTE doesn't communicate information but requires an intermediate element so it will be able to communicate.

DTE examples include computers, printer, routers and bridges, etc.

❑ Data Communications Equipment (DCE)

It is also called **Data Circuit-Terminating Equipment**. At the physical layer, the function of DCE is that it supplies that clocking. DCE provides the service. The DCE has devices whose function is to put data on the local loop.

also, as its function, **the DCE provides an interface for connecting subscribers to a communication link on the WAN cloud. Its function is to establish, maintain, and terminate communication network sessions between a data source and its destination.**

DCE examples include **ISDN adapters, Satellites (including base stations) Microwave stations, NIC (network interface cards) etc.**

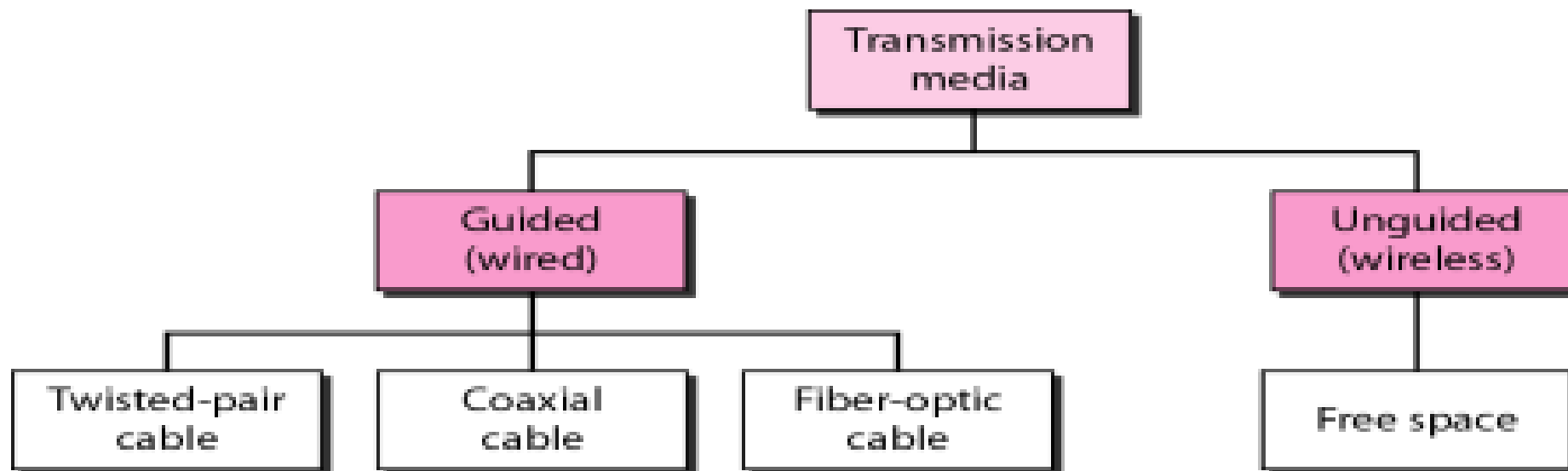
❑ DTE-DCE Interface

The major differences between DTE and DCE are as follows-

DTE	DCE
DTE is the acronym for Data Terminal Equipment	DCE is the acronym for Data Communications Equipment
DTE is to look for clocking.	DCE supplies that clocking.
Devices included in DTE are routers and computers.	Devices included in DCE are modem.
There should be no coordination between DTE devices.	Coordination should be required between DCE devices to communicate.
Examples - printers, computer, file servers.	Examples - Satellites, ISDN adapters, NIC.

❑ TRANSMISSION MEDIA

In data communication terminology, a transmission medium is a physical path between the transmitter and the receiver . it is the channel through which data is sent from one place to another. Transmission Media is broadly classified into the following types:



❑ Guided Media

It is also referred to as Wired or Bounded transmission media. Signals being transmitted are directed and confined in a narrow pathway by using physical links.

Features:

- High Speed
- Secure
- Used for shorter distances

There are 3 major types of Guided Media:

❑ Guided Media

Twisted Pair Cable : It consists of 2 separately insulated conductor wires wound about each other. Generally, several such pairs are bundled together in a protective sheath.

One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference. The receiver uses the difference between the two.

why aren't the two wires parallel?



❑ Twisted Pair Cable

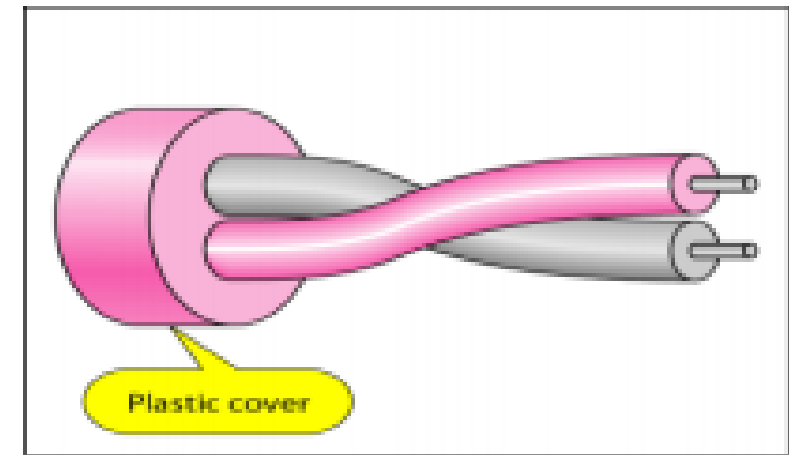
Unshielded Twisted Pair (UTP): This type of cable has the ability to block interference and does not depend on a physical shield for this purpose. It is used for telephonic applications.

Advantages:

- Least expensive
- Easy to install
- High speed capacity

Disadvantages:

- Susceptible to external interference
- Lower capacity and performance in comparison to STP
- Short distance transmission due to attenuation



a. UTP

❑ Twisted Pair Cable

Shielded Twisted Pair (STP):

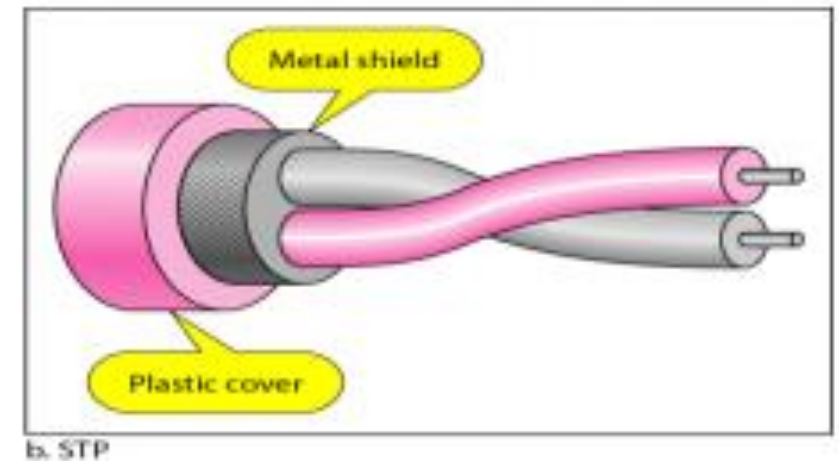
This type of cable consists of a special jacket to block external interference. It is used in fast-data-rate Ethernet and in voice and data channels of telephone lines.

Advantages:

- Better performance at a higher data rate in comparison to UTP
- Eliminates crosstalk
- Comparatively faster

Disadvantages:

- Comparatively difficult to install and manufacture
- More expensive
- Bulky



❑ Coaxial Cable

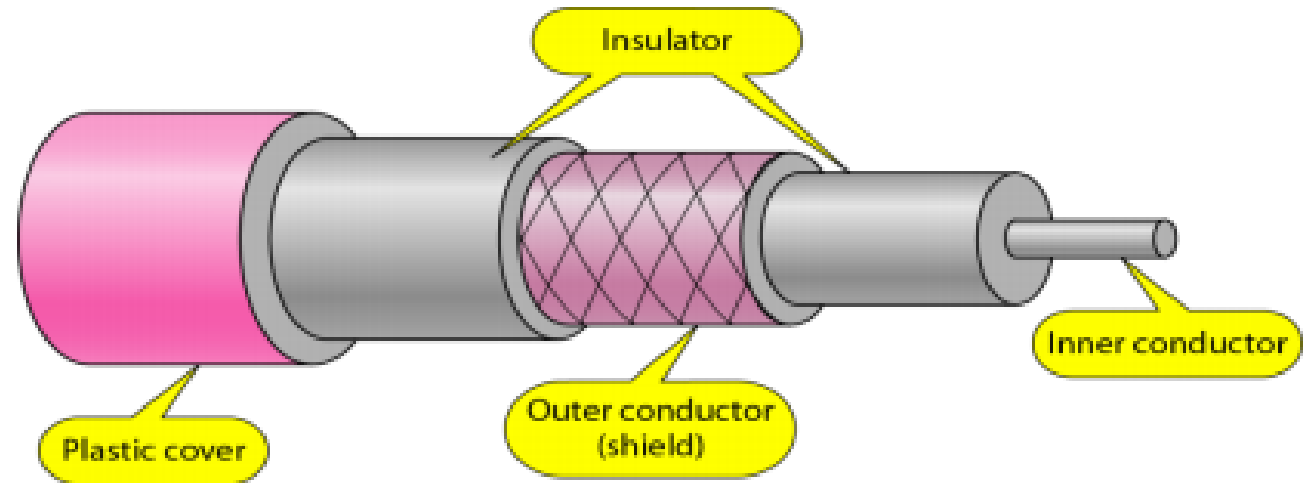
Coaxial cable (or coax) carries signals of higher frequency ranges than those in twisted pair cable. coax has a central core conductor of solid or stranded wire (usually copper) enclosed in an insulating sheath, which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two. The outer metallic wrapping serves both as a shield against noise and as the second conductor, which completes the circuit. This outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by a plastic cover. Cable TVs and analog television networks widely use Coaxial cables.

Advantages:

- High Bandwidth
- Better noise Immunity
- Easy to install and expand
- Inexpensive

Disadvantages:

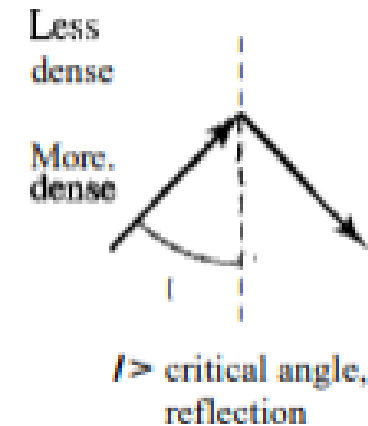
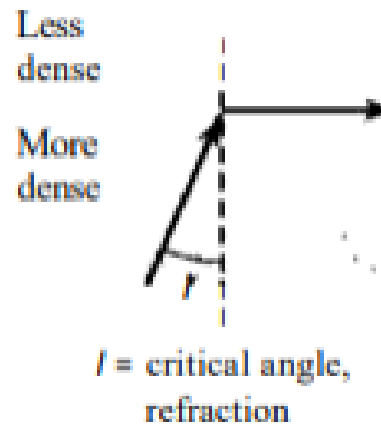
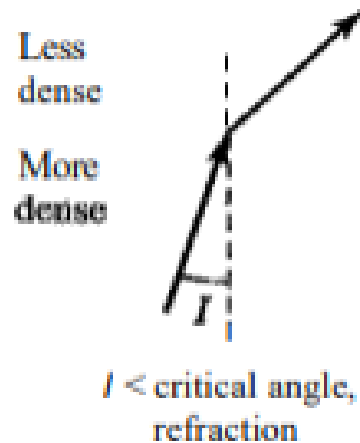
- Single cable failure can disrupt the entire network



□ Optical Fiber Cable

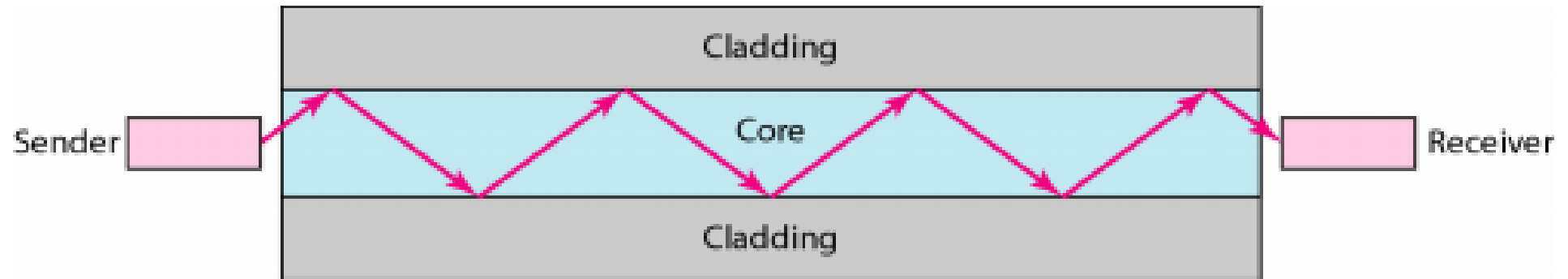
It uses the concept of **reflection of light** through a core made up of glass or plastic. The core is surrounded by a less dense glass or plastic covering called the cladding. It is used for transmission of large volumes of data.

A fiber-optic cable is made of glass or plastic and transmits signals in the form of light. To understand optical fiber, we first need to explore several aspects of **the nature of light**.



❑ Optical Fiber Cable

Optical fibers use reflection to guide light through a channel. A glass or plastic core is surrounded by a cladding of less dense glass or plastic. The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it.



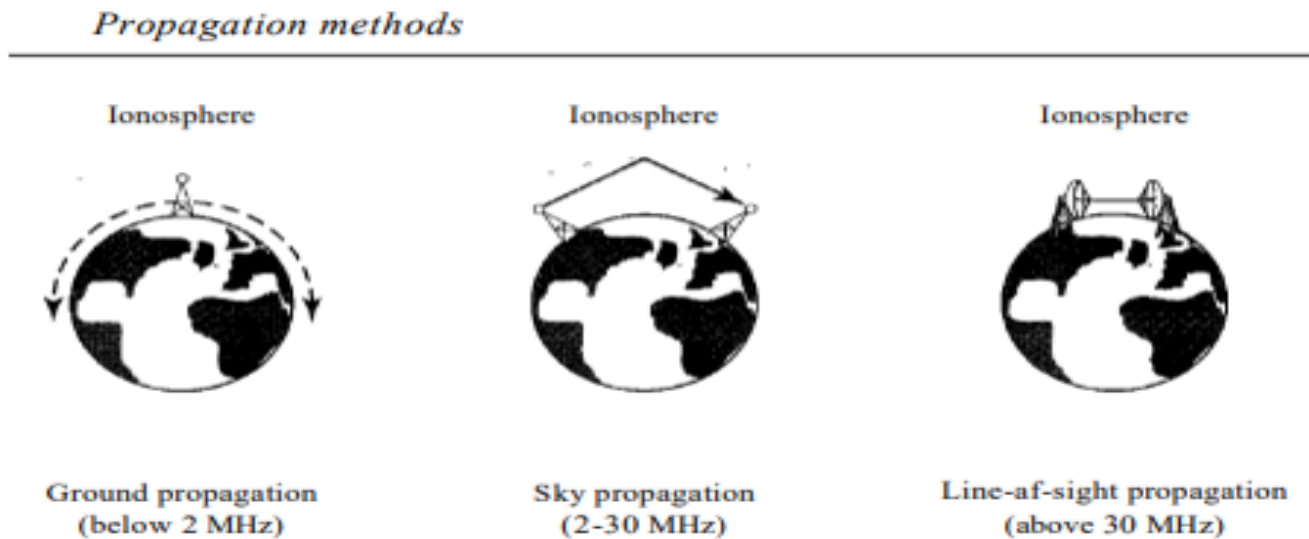
❑ Unguided Media

It is also referred to as Wireless or Unbounded transmission media. No physical medium is required for the transmission of electromagnetic signals.

Features:

- Signal is broadcasted through air
- Less Secure
- Used for larger distances

Unguided signals can travel from the source to destination in several ways: **ground propagation, sky propagation, and line-of-sight propagation.**



❑ Unguided Media

There are 3 major types of Unguided Media:

(1) Radio waves

Although there is no clear-cut demarcation between radio waves and microwaves, electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called radio waves; waves ranging in frequencies between 1 and 300 GHz are called microwaves.

Radio waves are omnidirectional. When an antenna transmits radio waves, they are propagated in all directions. This means that the sending and receiving antennas do not have to be aligned. A sending antenna sends waves that can be received by any receiving antenna.

Advantages

- Radio waves that propagate in the sky mode, can travel long distances. This makes radio waves a good candidate for long distance broadcasting such as AM radio. Radio wave
- can penetrate walls

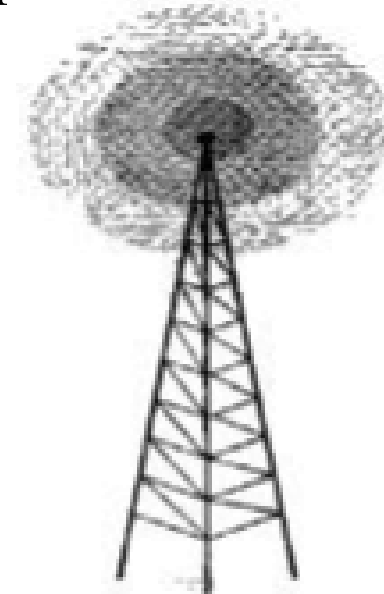
❑ Unguided Media

Dis advantages

- The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signals using the same frequency or band.
- The radio wave band is relatively narrow, just under 1 GHz, compared to the microwave band. When this band is divided into sub bands, the sub bands are also narrow, leading to a low data rate for digital communications.

Omnidirectional Antenna Radio waves use omnidirectional antennas that send out signals in all directions. Based on the wavelength, strength, and the purpose of transmission,

Applications The omnidirectional characteristics of radio waves make them useful for multicasting, in which there is one sender but many receivers. AM and FM radio, television, are examples of multicasting



❑ Unguided Media

(2) Micro wave

Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves. Microwaves are unidirectional. When an antenna transmits microwave waves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned.

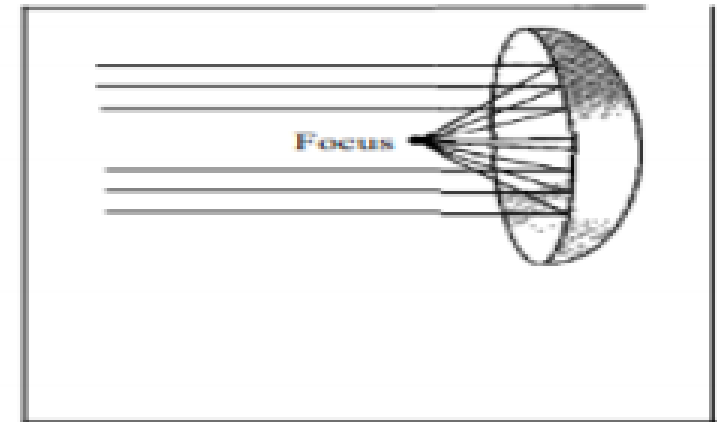
The following describes some characteristics of microwave propagation:

- Microwave propagation is **line-of-sight**. Since the towers with the mounted antennas need to be in direct sight of each other, towers that are far apart need to be very tall. as The curvature of the earth act as blocking obstacles not allow two short towers to communicate by using microwaves. **Repeaters are often needed for long distance communication.**
- Very high-frequency **microwaves cannot penetrate walls**. This characteristic can be a disadvantage **if receivers are inside buildings**
- The microwave band is relatively wide, **almost 299 GHz**. Therefore wider sub bands can be assigned, and a **high data rate is possible**
- Use of certain portions of **the band requires permission from authorities.**

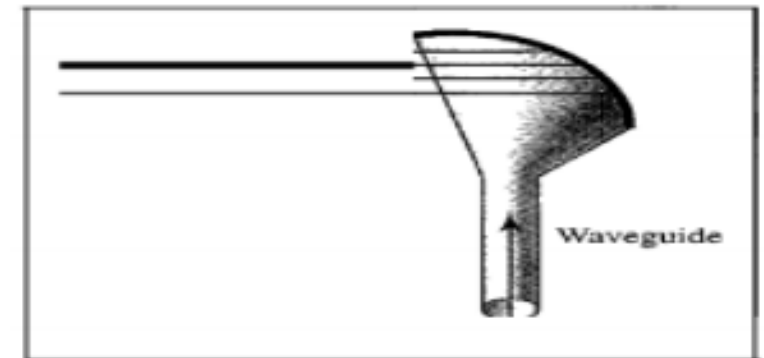
❑ Unguided Media

Unidirectional Antenna Microwaves need unidirectional antennas that send out signals in one direction. Two types of antennas are used for microwave communications: the parabolic dish and the horn (see Figure below). A **parabolic dish** antenna is based on the geometry of a parabola

A **horn antenna** looks like a gigantic scoop. Outgoing transmissions are broadcast up a stem and deflected outward in a series of narrow parallel beams by the curved head. Received transmissions are collected by the scooped shape of the horn, in a manner similar to the parabolic dish, and are deflected down into the stem



a. Dish antenna



b. Horn antenna

❑ Unguided Media

(3) Infrared

Infrared waves are used for very short distance communication. They cannot penetrate through

obstacles. This prevents interference between systems. It is used in TV remotes, wireless mouse, keyboard, printer, etc

➤ **infrared waves**, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication. Infrared waves, having high frequencies, cannot penetrate walls. This advantageous characteristic prevents interference between one system and another; a short-range communication system in one room cannot be affected by another system in the next room.

When we use our infrared remote control, we do not interfere with the use of the remote by our neighbors. However, this same characteristic makes infrared signals useless for long-range communication. In addition, we cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication.

Thanks

❑ quiz2

1. Which address identifies a specific process on a host?
a) MAC b) IP c) Port d) Domain
2. Which protocol provides reliable delivery in TCP/IP?
a) UDP b) IP c) TCP d) ARP
3. Which layer of TCP/IP corresponds to the Transport layer in OSI?
a) Application b) Internet c) Transport d) Network Access
4. Which layer provides end-to-end communication and error recovery?
a) Network b) Transport c) Session d) Physical
5. Which OSI layer converts bits into signals for transmission?
a) Application b) Transport c) Data Link d) Physical